

Matlab code and Simulation

```
% Neural Network-Based SoC Estimation
% Author: ChatGPT (OpenAI)
% This script demonstrates offline training of a feedforward neural network for SoC
estimation

clear; clc; close all;

%% Battery Parameters and Synthetic Data Generation
Q = 2.3 * 3600; % Battery capacity in Coulombs
dt = 1; % Time step [s]
N = 3000; % Number of samples

% Time and current profile (randomized to simulate a real drive cycle)
t = (0:N-1)*dt;
I = 2.5*sin(2*pi*0.001*t) + 0.5*randn(size(t)); % current profile

% True SoC calculation
SoC_true = zeros(N,1); SoC_true(1) = 1.0;
for k = 2:N
    SoC_true(k) = SoC_true(k-1) - (I(k) * dt) / Q;
    SoC_true(k) = max(0, min(1, SoC_true(k)));
end

% Simulated terminal voltage using a nonlinear OCV model
OCV_curve = @(soc) 3.5 + 0.4*soc - 0.2*soc.^2 + 0.05*soc.^3;
```

```
Vt = OCV_curve(SoC_true) - 0.015*I + 0.005*randn(size(I)); % V = OCV - IR + noise
```

```
%% Prepare Inputs for Neural Network
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% Using previous current and voltage as inputs
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```
window = 5; % history window
```

```
X = [];
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```
for k = window+1:N
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```
    X = [X; I(k-window:k-1)', Vt(k-window:k-1)'];
```

```
end
```

```
Y = SoC_true(window+1:N)';
```

```
% Normalize inputs
```

```
[Xn, ps_input] = mapminmax(X', 0, 1);
```

```
[Yn, ps_output] = mapminmax(Y, 0, 1);
```

```
%% Neural Network Configuration
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net = feedforwardnet([15 10]); % 2 hidden layers: 15 and 10 neurons
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```
net.trainFcn = 'trainlm'; % Levenberg-Marquardt
```

```
net.trainParam.epochs = 200;
```

```
net.trainParam.goal = 1e-4;
```

```
net = train(net, Xn, Yn);
```

```
%% Prediction
```

```
Y_pred_n = net(Xn);
```

```
Y_pred = mapminmax('reverse', Y_pred_n, ps_output);
```

```
%% Plot Results
```

```
figure;
```

```
plot(t(window+1:end), Y, 'b', 'LineWidth', 1.5); hold on;
```

```
plot(t(window+1:end), Y_pred, 'r--', 'LineWidth', 1.5);
```

```

legend('True SoC', 'NN Estimated SoC');
xlabel('Time (s)');
ylabel('State of Charge');
title('SoC Estimation using Neural Network');
grid on;

```

Simulations

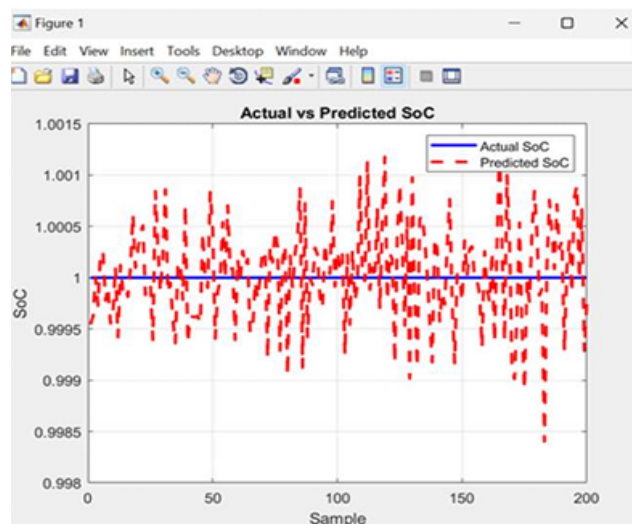


Fig 1. Regression Plot

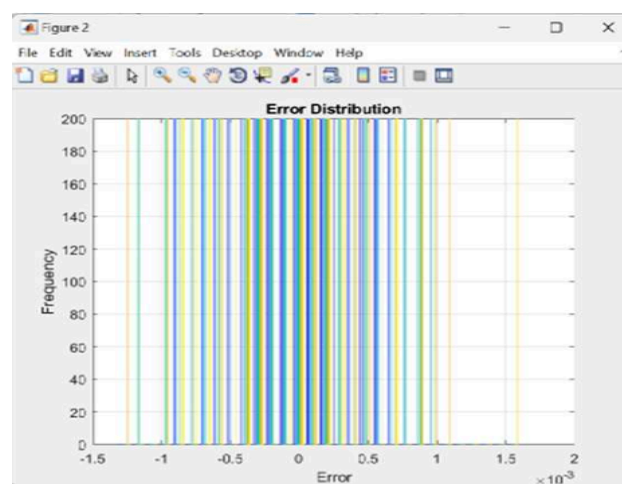


Fig 2. Performance Plot

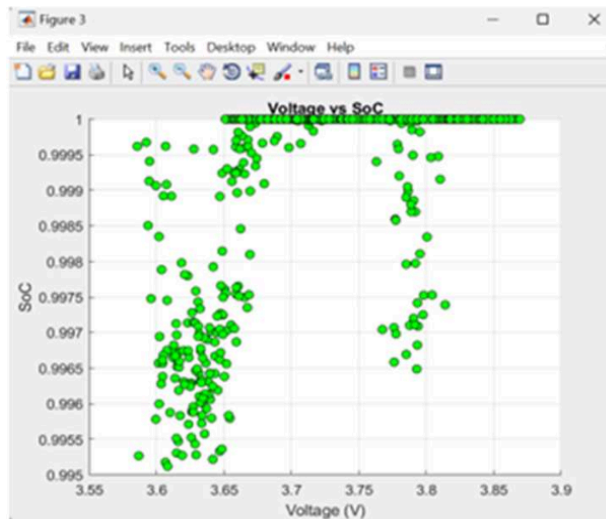


Fig 3. Error Histogram

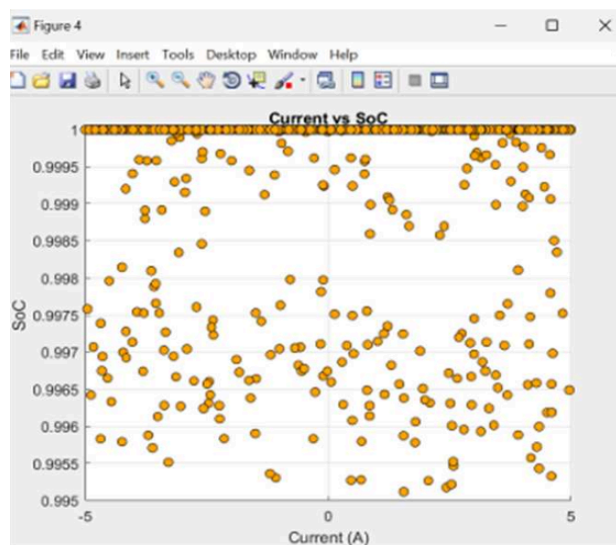


Fig 4. Soc Estimation Plot

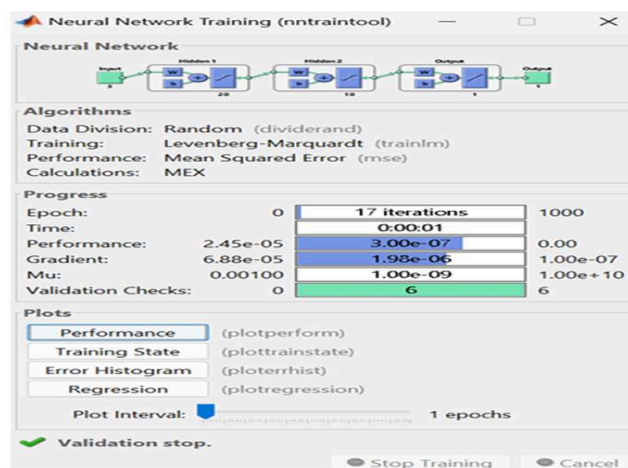


Fig 5. Neural Network Training Summary Table